Research Participation Report

Solar PV Output Prediction and Ensemble Evaluation

Using Plant Records and Korea Meteorological Administration Data

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1. Research Overview

During my research participation at POSTECH SPaC Lab from January to February 2026, I implemented and compared regression models that estimate solar photovoltaic (PV) output from sensor and weather observations recorded at the same time. I first examined the structure of the SQLite data and merged the sensor and inverter records. I then calculated Pearson correlation coefficients between the output and input variables to select the baseline features, and trained Random Forest, LightGBM, and XGBoost models.

To supplement the weather measurements from the plant's sensors, I conducted an experiment that incorporated air temperature, wind speed, and relative humidity data from the Korea Meteorological Administration (KMA). I also implemented an ensemble that equally averages predictions from three separately trained and saved models, and compared it with the individual models. This report describes data processing, feature selection, experimental settings, model performance, and limitations in interpreting the results.

The plant operating data were provided by an external source. My work covered conversion from SQLite to CSV, inspection of duplicate records, timestamp-based data merging, feature construction, and implementation of model training and evaluation programs. Among the five plants in the supplied dataset, the Yangsan plant was selected because its records were the most complete.

2. Data-Driven Solar PV Output Prediction

2.1 Data Sources and Study Site

The supplied SQLite database contained sensor and inverter operating records. After exporting the database tables to CSV files with column names, I merged the sensor and inverter records using the measurement timestamp, stats_date.

Dataset	Rows	Contents
Raw sensor records	131,856	Timestamp, sensor ID, irradiance, ambient temperature, and surface temperature
Raw inverter records	131,695	Timestamp, voltage, current, output, and operating status
Merged data after sensor selection	131,693	Timestamp-based merge after selecting sensor_id = 990

The raw sensor data contained duplicate timestamps. Selecting rows with sensor_id = 990 reduced the number of duplicate timestamps from 89 to 0. The selected sensor records were then merged with the inverter records for subsequent analysis.

2.2 Feature Selection and Data Preprocessing

PV_gen_power was used as the target variable, and Pearson correlation coefficients were calculated for the numerical variables. The initial analysis considered voltage, current, output, and weather data before selecting irradiance and temperature variables as features. The table below summarizes the correlations between the main weather variables and output in the merged dataset.

Variable	Pearson correlation with PV_gen_power	Baseline feature
Horizontal irradiance	0.9438	Included
Tilted irradiance	0.7194	Included
Surface temperature	0.5785	Included
Ambient temperature	0.1790	Excluded

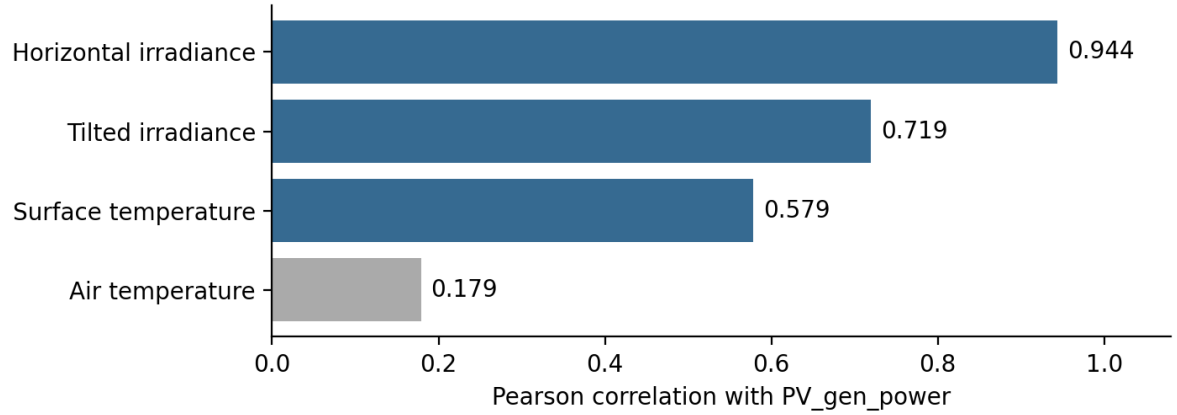


Figure 1. Pearson correlations between the baseline weather variables and PV_gen_power.

The baseline experiment used horizontal irradiance, tilted irradiance, and surface temperature, excluding ambient temperature measured at the plant. The Pearson correlation between ambient temperature and PV_gen_power was approximately 0.179, lower than those of the other three variables. However, this value alone does not establish that ambient temperature makes no contribution to prediction by a nonlinear model.

The final ensemble used seven features prepared through a separate training workflow, including ambient temperature. The independent effect of adding ambient temperature was not evaluated in this report.

Column or group	**Treatment**
data_index, sensor_id, inverter_id	Used to identify records or devices; excluded from model inputs
stats_date	Used for data merging and time filtering; excluded from model inputs
Voltage, current, generated_power, etc.	Excluded from the baseline model inputs
total_generated_power, line_power_loss, line_freq, ready	Excluded from the final feature list
stop	Used to remove records with stopped operation, then excluded from model inputs

Preprocessing removed rows indicating stopped operation (stop = 1) and retained observations from 07:00 to 17:00. The data were averaged at hourly intervals, missing values were removed, and only dates with all 11 hourly observations were retained. The final baseline dataset contained 15,917 samples over the valid observation periods between April 13, 2018 and January 6, 2023.

The features and target were min–max normalized to the range [0, 1]. The stop column was removed from the feature list after filtering.

2.3 Integration of KMA Weather Data

KMA data were added to supplement the plant's weather measurements. The KMA timestamp column, tm, was renamed stats_date, and the date and time formats were standardized before merging records with matching timestamps. The selected KMA features were air temperature (ta), wind speed (ws), and relative humidity (hm).

Experiment	Features	Target
Baseline	Horizontal irradiance, tilted irradiance, and surface temperature	PV_gen_power
With KMA data	The three baseline features plus KMA air temperature, wind speed, and relative humidity	PV_gen_power

The experiment with KMA data also applied operating-status and time filtering, hourly averaging, missing-value removal, selection of dates with 11 samples, and min–max normalization. The final dataset contained 10,934 samples from April 15, 2018 to November 9, 2022. Its sample count and observation period differed from those of the baseline experiment, and the restriction to sensor_id = 990 was not applied in the same way.

2.4 Training and Evaluation of Three Models

Both the baseline experiment and the experiment with KMA data used three regression models: Random Forest, LightGBM, and XGBoost. Random Forest aggregates predictions from multiple decision trees. LightGBM and XGBoost are boosting models that sequentially train trees to reduce errors in earlier predictions. Each trained model was used to predict output for the test data.

Model	Main training settings
Random Forest	n_estimators = 100; random_state = 1
LightGBM	n_estimators = 300; learning_rate = 0.05; random_state = 1
XGBoost	n_estimators = 300; learning_rate = 0.05; random_state = 1

In both experiments, the data were randomly shuffled and split into 80% training data and 20% test data. Performance was evaluated using the coefficient of determination (R^2) and root mean squared error (RMSE). On the same test data, R^2 values closer to 1 and RMSE values closer to 0 indicate better predictive performance. RMSE in these two experiments was calculated using the min–max-normalized target.

The final ensemble evaluation used separately trained and saved models. In that workflow, generated_power was normalized by the plant capacity, and the data were split 80:20 into training and test sets without shuffling. As summarized below, the three experiments differed in their targets, features, normalization methods, and data splitting procedures. Their

performance values therefore cannot be directly compared as results from an identical experimental setup.

The final model used future_step = 0, meaning that it estimated output at the same time as the input observations. This differs from forecasting the next day's output using future weather forecasts.

Experiment	Target	Features	Normalization and split
Baseline	PV_gen_power	3 features: horizontal and tilted irradiance, surface temperature	Min–max normalization; random 80:20 split
With KMA data	PV_gen_power	6 features: baseline features plus KMA air temperature, wind speed, and relative humidity	Min–max normalization; random 80:20 split
Final ensemble	generated_power	7 features: horizontal and tilted irradiance, surface and ambient temperatures, and the three KMA variables	Plant-capacity normalization; 80:20 split preserving record order

2.5 Performance Comparison With and Without KMA Data

The results of the baseline experiment and the experiment with KMA data are summarized below. All three models achieved higher R^2 and lower RMSE in the experiment using KMA data. LightGBM performed best among the three models in that experiment, with R^2 of approximately 0.9379 and RMSE of approximately 0.06329.

Model	Baseline R^2	With KMA R^2	Baseline RMSE	With KMA RMSE
Random Forest	0.8959	0.9365	0.08301	0.06396
LightGBM	0.9007	0.9379	0.08108	0.06329
XGBoost	0.9007	0.9377	0.08105	0.06335

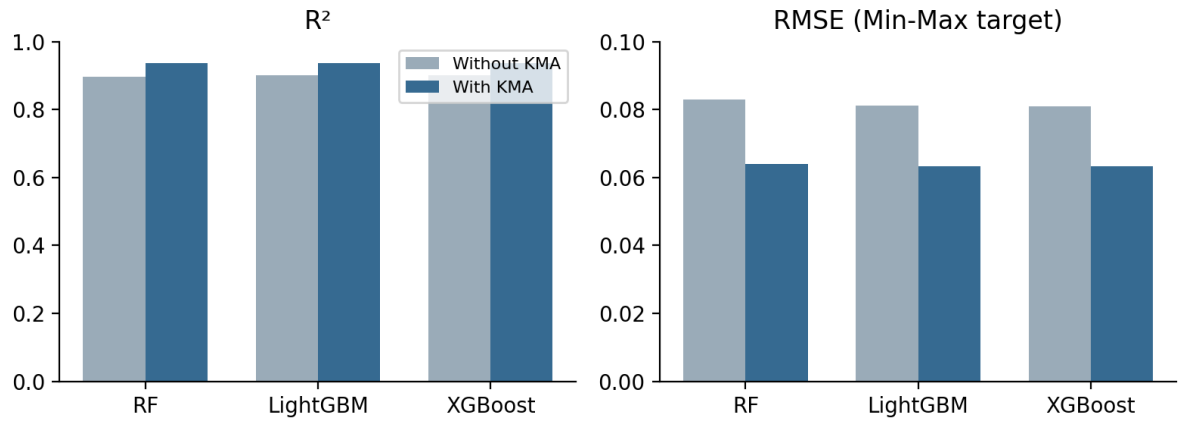


Figure 2. Model performance in the baseline experiment and the experiment with KMA data.

Improved performance was observed in the experiment using KMA data. However, the two experiments differed in sample count, observation period, and some preprocessing conditions. The random_state argument in train_test_split was also not fixed, and min–max normalization was performed before the training/test split. Consequently, the performance differences cannot be attributed entirely to the addition of KMA variables; they should be interpreted as results obtained under the respective recorded experimental settings.

To evaluate the effect of KMA data more precisely, the experiments should use identical test samples from common dates and vary only the input features. The min–max normalization parameters should also be calculated using only the training data and then applied to the test data.

2.6 Ensemble Construction and Final Evaluation

An equally weighted ensemble was constructed to combine the predictions of the three models. The Random Forest, LightGBM, and XGBoost models trained on the Yangsan data were loaded, and their test inputs, observed outputs, and dates were checked for alignment. The three predictions were then averaged with equal weights.

$$\text{Ensemble prediction} = (\text{Random Forest prediction} + \text{LightGBM prediction} + \text{XGBoost prediction}) / 3$$

The final ensemble prediction was calculated as the arithmetic mean of the three model outputs, without retraining the individual models.

The final evaluation used all 3,602 test samples from the Yangsan plant. The features were horizontal irradiance, tilted irradiance, ambient temperature, surface temperature, and KMA wind speed, air temperature, and relative humidity. The target was generated_power. The RMSE values below were calculated for output normalized by plant capacity and therefore cannot be directly compared with the min–max-normalized RMSE values in the preceding section.

Model	R^2	RMSE (capacity-normalized target)
Random Forest	0.9306	0.0675
LightGBM	0.9340	0.0659
XGBoost	0.9312	0.0673
Ensemble	0.9360	0.0649

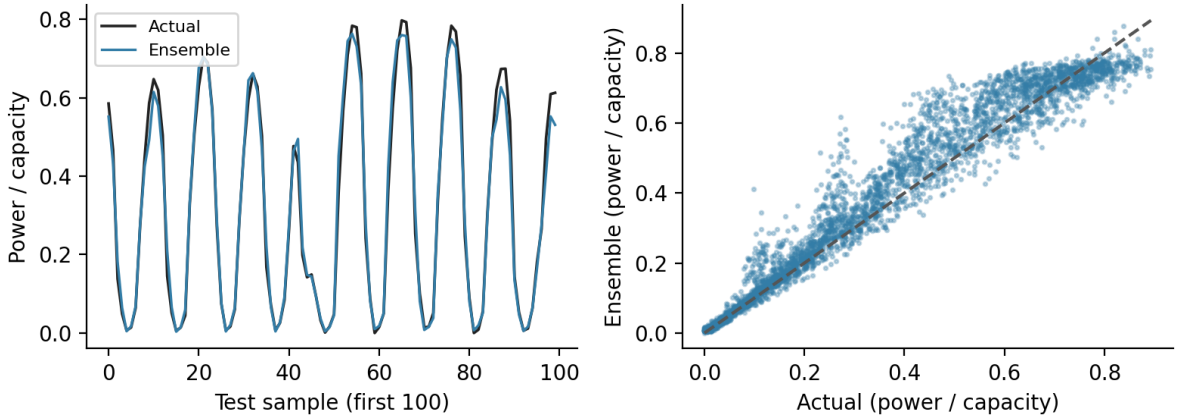


Figure 3. Observed output and ensemble predictions for the Yangsan test data.

On the same test data, the ensemble achieved R^2 of 0.9360 and RMSE of 0.0649. Compared with LightGBM, the best individual model (R^2 = 0.9340; RMSE = 0.0659), the ensemble increased R^2 by 0.0020 and reduced RMSE by 0.0010. Although the improvement was modest, averaging predictions was slightly more effective than using the individual models under these evaluation conditions.

The left panel of Figure 3 compares the observed and predicted values for the first 100 test samples, while the right panel shows a scatter plot for all test samples. The horizontal axis in the left panel represents sample order, not a continuous time axis.

3. Conclusion

During this research participation, I converted operating records from the Yangsan solar PV plant into a form suitable for analysis and merged sensor and inverter records by timestamp. After resolving duplicate records, I constructed the baseline feature set using Pearson correlation coefficients and implemented the training and evaluation of Random Forest, LightGBM, and XGBoost models. I also integrated KMA data and evaluated an ensemble that combined predictions from the saved individual models.

The experiment with KMA data showed higher R^2 and lower RMSE for all three models. However, because its samples, observation period, and preprocessing conditions differed from those of the baseline experiment, these differences cannot be interpreted as the independent effect of the KMA variables. On identical test data, the ensemble achieved R^2 of 0.9360 and RMSE of 0.0649, representing a modest improvement over the individual models.

The models in this study estimated output at the same time as the input observations; their ability to forecast future output was not evaluated. Future comparisons should use a fixed set of test samples, split the data with temporal order taken into account, and calculate normalization parameters using only the training data to ensure consistent evaluation conditions.